

Price Range Estimation for Seattle Airbnb Entire-Unit Listings: A Structured Modeling Approach

1. INTRODUCTION

Setting an appropriate nightly price is one of the most consequential decisions a new Airbnb host must make. Pricing too high reduces booking volume and suppresses guest reviews, while pricing too low erodes revenue and may signal lower perceived quality. Yet the platform provides no structured guidance calibrated to Seattle market conditions, leaving first-time hosts to rely on informal comparisons or trial-and-error.

This analysis builds a predictive pricing model for entire-unit listings in Seattle to offer new hosts a data-driven price range for their property. The model draws on publicly available listing data and encodes the structural features that the market consistently rewards or penalizes. Given the inherent heterogeneity of Airbnb inventory, including differences in decor, furnishings, and host presentation that no structured dataset can fully capture, an expected Mean Absolute Percentage Error (MAPE) of roughly 20 to 25 percent is appropriate and consistent with published benchmarks for short-term rental pricing models. The practical implication is that model outputs should be interpreted as a probable price band rather than a precise point estimate.

2. DATA AND METHODOLOGY

2.1 Data Source and Preprocessing

The dataset was obtained from Inside Airbnb, and originally contained 6997 Seattle listings scraped in August 2025. Two duplicate rows were removed, and observations missing a review score rating were dropped. Remaining missing values in numeric columns (bathrooms, bedrooms, beds, host lifetime, and price) were imputed using a kNN imputer. Categorical variables with sparse missingness were imputed using the mode or median as appropriate.

2.2 Sample Restrictions: Minimum and Maximum Nights

Two night-stay filters were applied before modeling, each reflecting a distinct aspect of the Seattle short-term rental (STR) market.

For maximum nights, observations were retained only if the listing cap was set at or below 365 nights. Seattle enacted STR regulations that restrict non-primary-residence rentals to relatively few nights per year, with 30 nights as limit. However, compliance in practice is low: in the raw dataset, 25 percent of listings set a maximum-nights cap of 365 or higher, indicating that the modal host may treat 365 as a default setting rather than an active policy choice. Retaining observations at or below 365 nights therefore excludes extended-stay listings, which operate on a fundamentally different pricing logic resembling long-term rentals, while keeping the vast majority of vacation-rental hosts in the sample regardless of their formal STR compliance status.

For minimum nights, observations were restricted to those requiring 14 or fewer consecutive nights. This threshold separates vacation rentals since extended-stay or medium-term rentals attract a different guest segment and command structurally different prices. Additionally, property owners in Seattle may need extra paper work if extended stay is initiated.

2.3 Scope: Entire-Unit Listings and Outlier Removal

The analysis focuses exclusively on entire home or apartment listings. Private and shared room listings were excluded because their pricing dynamics differ from entire-unit stays. Top-one-percent price outliers (listings priced above approximately \$722 per night) were also removed to prevent a small number of luxury properties from distorting model training. After all filters, the final analytic sample contained 3,255 listings with nightly prices ranging from \$49 to \$722.

2.4 Feature Engineering

Amenity coverage was captured in two ways: a count of all listed amenities (ranging from 5 to 90 per listing, with a mean of 46) and binary indicators for specific featured amenities including hot tub, WiFi, stove, free street parking, workspace, TV, and washer. Neighborhood geography was consolidated from 17 distinct Seattle neighborhoods into three groups: Downtown Seattle (Downtown core, Cascade, Capitol Hill, Queen Anne), North Seattle (Ballard, Interbay, Lake City, Magnolia, Northgate, University District), and Southwest/Others. A binary indicator flagged listings that received a review in 2025, serving as a proxy for recent booking activity. Host response time was recoded into a fast-response indicator (response within one hour). Host lifetime in years was calculated from the account creation date to the scrape date.

2.5 Outcome Variable and Model Features

The nightly price was log-transformed using $\ln(1 + \text{price})$ to reduce right skewness and stabilize variance. The final feature set entering all models consisted of 20 predictors: host acceptance rate, superhost status, bathrooms, bedrooms, accommodates, 30-day forward availability, overall review score, cleanliness review score, estimated occupancy over the trailing 365 days, neighborhood group dummies, recent review indicator, total amenity count, and binary amenity indicators.

3. EXPLORATORY DATA ANALYSIS

3.1 Price Distribution

Before filtering, the data showed a highly right-skewed price distribution with a mean of \$181 and a maximum exceeding \$9,400, either reflecting a small number of luxury estate properties or incorrect data entries. After restricting to entire-unit listings and removing the top one percent, the distribution became more manageable: the median nightly price was \$164, the mean was \$186, the interquartile range spanned \$123 to \$222, and the maximum was \$722. The residual right skew confirmed the appropriateness of a log transformation for regression modeling. Figure 1 illustrates both the price distribution and the variation across neighborhood groups.

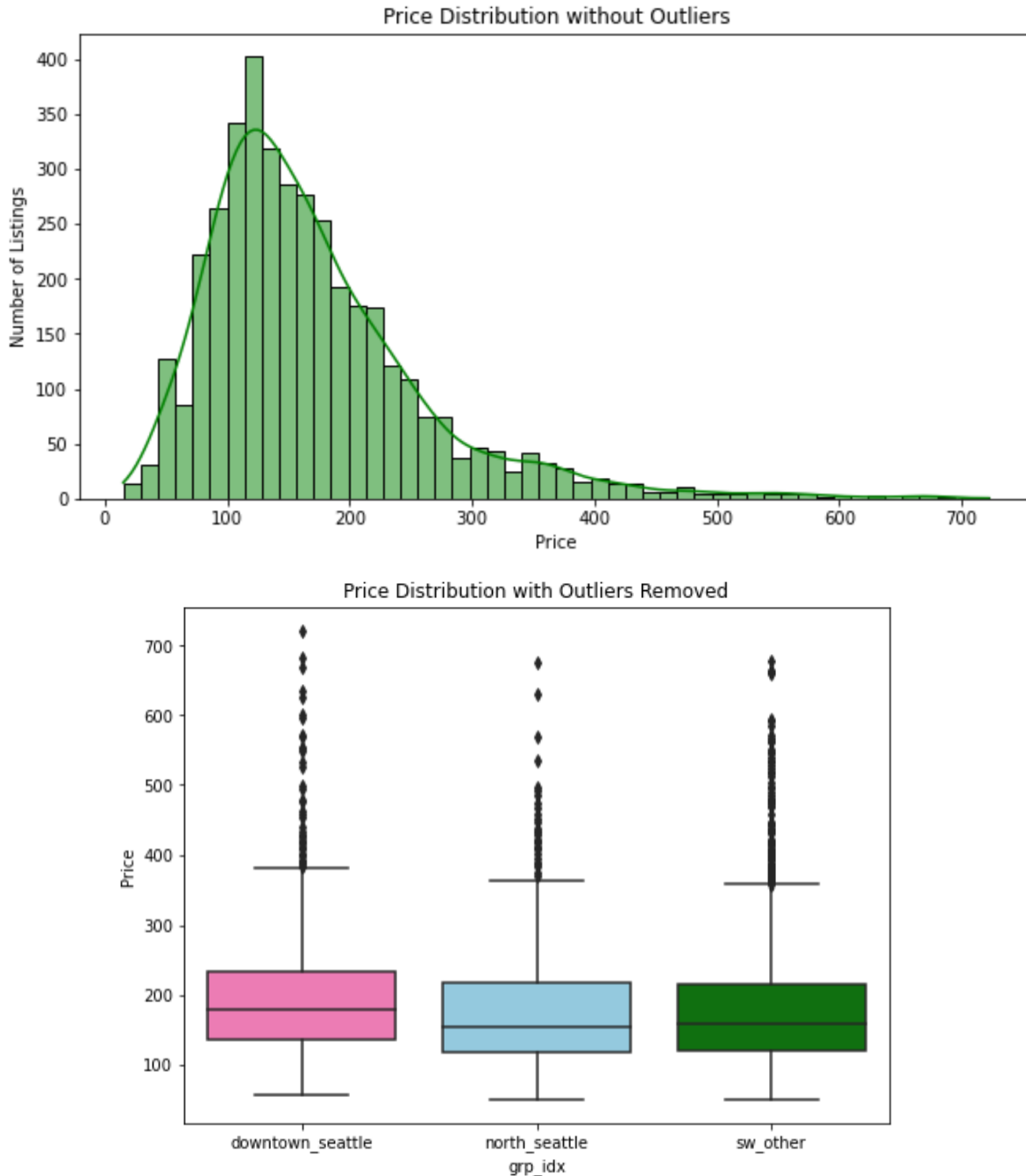


Figure 1. Left: Nightly price distribution of entire-unit Seattle listings after outlier removal ($n = 3,929$); median (\$164) and mean (\$186) shown in dashed lines. Right: Price distribution by neighborhood group; downtown listings command the highest median.

3.2 Neighborhood Geography

Downtown Seattle commanded the highest median prices, consistent with its centrality and proximity to tourism and convention infrastructure. North Seattle listings posted lower median prices with somewhat less dispersion. Southwest Seattle and other neighborhoods, which constitute the largest share of listings by count, exhibited the widest price spread. These geographic

patterns are later confirmed by the large, statistically significant neighborhood coefficients in the regression model.

3.3 Amenities and Host Characteristics

The average listing offered 46 amenities. WiFi was nearly universal (present in over 99 percent of listings), as was a washer (88 percent). Hot tubs were present in only 9 percent of listings, making them a potential differentiator. Free street parking was available for roughly 70 percent of listings. The host acceptance rate was high on average (mean 0.96), and 93 percent of hosts responded within one hour. Approximately 95 percent of listings had received a review in 2025.

3.4 Availability and Occupancy

Downtown Seattle listings had lower forward availability, consistent the neighborhood being the top reviewed neighborhood in the previous years, signaling high demand. The estimated trailing 365-day occupancy variable showed a negative bivariate association with price, suggesting a high-volume / lower-price segment. It is also worth noting that availability and occupancy did not fully demonstrate booking demand as they are either blocked by host or genuinely booked by renters.

4. MODEL EVALUATION

4.1 Models Estimated

Four regression models were estimated on the log-transformed price: Ordinary Least Squares (OLS), Random Forest, XGBoost, and a linear-kernel Support Vector Regression (SVR). All models used an identical feature set and train-test split. Table 1 summarizes out-of-sample performance on the held-out 30 percent test set.

Table 1. Model Performance Comparison (Test Set, n = 977)

Model	R ² (Log)	R ² (Org.)	RMSE (Org.)	MAE (Org.)	MAPE
OLS	0.614	0.582	57.70	39.35	0.205
Random Forest	0.593	0.559	59.29	40.25	0.210
XGBoost	0.557	0.494	63.53	41.70	0.215
SVR	0.604	0.584	57.62	39.51	0.207

Note: Bold values indicate the best-performing model for each metric. RMSE and MAE are in USD.

4.2 Post-Estimation Correction for OLS

The initial OLS model was estimated with conventional homoscedastic standard errors, yielding an R-squared of 0.614 on the log scale. Visual inspection of the residuals versus fitted values plot revealed heteroscedasticity. To produce reliable inference, the model was re-estimated with Huber-White HC3 heteroscedasticity-consistent standard errors. The metrics in Table 1 correspond to this HC3 specification.

4.3 OLS Outperforms XGBoost

XGBoost ranked last among the four models on every metric, with a MAPE of 21.5 percent and an original-scale R-squared of 0.494. In this environment, ensemble complexity yields minimal benefit. Even though XGBoost configuration was already tuned (shallow depth of three, low learning rate of 0.02, subsampling of 0.70), the model could not match the simpler linear specification. Some hypothesized reason for the outperformance of OLS include : First, the log transformation potentially linearized the price-feature relationship, directly benefiting OLS; secondly, the feature set was constructed from variables known on theoretical grounds to influence pricing, which may align well with OLS assumptions. SVR with a linear kernel achieved performance nearly identical to OLS (MAPE 20.7 percent versus 20.5 percent), potentially reinforces that a linear decision boundary is well suited to this transformed dataset. In this analysis, OLS is preferred over SVR as it yields interpretable coefficients that a new host can directly translate into pricing intuition.

4.4 Robust OLS Results: Significant Predictors

Table 2 presents all predictors from the HC3 robust OLS model that are statistically significant at the five percent level. The dependent variable is $\ln(1 + \text{price})$, so coefficients approximate the percentage change in price for a one-unit increase in the predictor, holding all else constant.

Table 2. Robust OLS Results with Significant Variables (HC3 Standard Errors)

Variable	Coef.	Std. Err.	z	P> z	[0.025	0.975]
constant	4.008	0.274	14.62	0.000	3.470	4.545
host_acceptance_rate	-0.152	0.076	-2.01	0.045	-0.301	-0.004
bathrooms	0.121	0.014	8.72	0.000	0.094	0.148
bedrooms	0.127	0.012	10.59	0.000	0.103	0.150
accommodates	0.022	0.005	4.14	0.000	0.011	0.032
availability_30	0.002	0.001	2.74	0.006	0.001	0.004
eeview_score_cleanliness	0.165	0.049	3.37	0.001	0.069	0.261
estimated_occupancy_l365d	-0.001	0.000	-10.34	0.000	-0.001	-0.001
north_seattle	-0.166	0.019	-8.92	0.000	-0.203	-0.130
sw_others	-0.176	0.014	-12.37	0.000	-0.204	-0.148
recent_reviewed	-0.075	0.037	-2.03	0.043	-0.147	-0.002
amenity_count	0.005	0.001	7.31	0.000	0.004	0.006
free_street_parking	-0.059	0.015	-3.99	0.000	-0.089	-0.030
hot_tub	0.095	0.019	4.87	0.000	0.057	0.133
wifi	-0.368	0.107	-3.43	0.001	-0.579	-0.158

Note: Dependent variable is $\ln(1 + \text{price})$. $n = 2,278$ (training set). Standard errors are HC3 heteroscedasticity-consistent. Reference categories: Downtown Seattle (neighborhood), not recently reviewed.

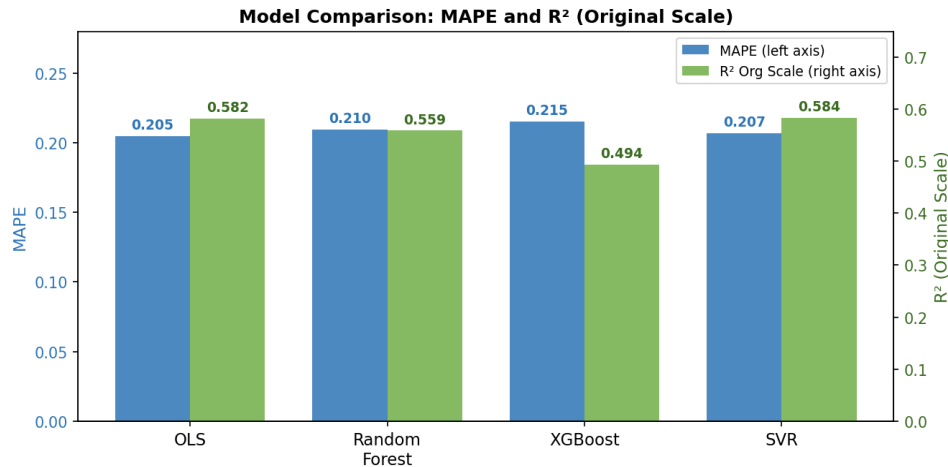


Figure 2. MAPE (left axis) and R^2 on original price scale (right axis) for all four models. OLS achieves the lowest MAPE; XGBoost underperforms materially.

4.5 Interpretation of Key Coefficients

Physical capacity predictors dominate the coefficient table. Each additional bathroom is associated with a 12.1 percent price premium, and each additional bedroom with a 12.7 percent premium. Accommodation capacity adds a further 2.2 percent per additional guest.

Location premiums are substantial. Being outside Downtown Seattle carries a discount of 16.6 percent for North Seattle and 17.6 percent for Southwest and other areas. Among amenities, the hot tub premium stands out at 9.5 percent, confirming that functionally rare amenities justify meaningful price increases. The WiFi coefficient is negative and large (-36.8 percent), this must be interpreted with caution. Free street parking carries a statistically significant negative coefficient (-5.9 percent). A high host acceptance rate is negatively associated with price, likely suggesting that hosts accepting all comers tend to operate more volume-oriented, budget-friendly listings. Higher trailing occupancy is negatively associated with price, potentially indicating a high-volume/discounted segment distinct from premium, lower-occupancy listings.

5. CONCLUSION

This analysis provides a data-driven framework for new Airbnb hosts in the Greater Seattle area to set their nightly pricing decisions. The most actionable findings for a new host are: physical capacity (bedrooms, bathrooms, and guest count) is the primary price determinant and should anchor initial rate setting; downtown location commands roughly a 17 percent premium over comparable non-downtown listings; distinctive amenities, particularly a hot tub, support meaningful premium pricing; and competitive cleanliness review scores matter for sustaining price over time.

The robust OLS model explains 61 percent of the variation in log-transformed price and achieves a MAPE of approximately 20.5 percent, consistent with the nature of Airbnb pricing driven by unobservable listing quality and demand shocks. Model outputs are best used as a probable price

band: a new host should treat the model prediction as the midpoint of a roughly 20-percent confidence interval and then adjust based on qualitative and seasonality factors.

Additionally, some limitations worth addressing is the dataset is a single cross-sectional snapshot from August 2025, likely capturing a seasonal peak, so estimates may not generalize reliably to off-season pricing. Unlike the hotel industry, where verified occupancy rates are reported, Airbnb does not disclose true booking data; the occupancy variable used here is a modeled proxy derived from listing calendar rather than actual reservation records, introducing measurement error into one of the stronger predictors and limiting the model's ability to fully account for demand-side pricing signals. Furthermore, the absence of unobservable listing quality factors, including photography, and listing description quality, may prevent structured-data models from achieving MAPE below roughly 20 percent. Future work could address these gaps through incorporating NLP-derived features from listing text to capture unobservable quality signals, potentially reducing MAPE to the level below 20 percent.